

Ple Encryption

OPERATION 0xCYNX

Challenge Author: Medo

Category: Cryptography

Difficulty: Medium

Classification: Official

Description

You are tracking down a secret organization. With the help of the newly developed AI Assistant, you gather enough information to track them down. Can you decrypt their conversation and retrieve the flag?

Synopsis

Determine the **encoding** method used in the chat logs and attempt to write a **decoder** to decipher their conversation and recover the password to the Archive.

Analysis

Running the executable, we are greeted with an “AI” that provides us with some files. The files we are given are “**Recover.ed.txt**”, “**Archive.ed.txt**”, and “**Archive.7z**”. Looking at the text files, we notice that it’s a conversation between an unknown person and someone called Ford Pines.

--- Original Message ---

From: [REDACTED]
Sent: Friday, June 15, 2012 18:34
To: FrPines@Mystery.org
Subject: PIe Recipe

>

Hello Mr. Ford,

I have found a PIe recipe. I would love for you to try it, but I don't want anyone knowing the recipe. So I'll cipher the recipe, so that no one will be able to read it.

V28gZGt4aSBsemxzIEdTciwgY2dtIGthbW8gZ3MgeXd3IHR5byBzc2Rkcnd6eHQgbWZ5dWV1c3JybGsuDQoxLiB3bGZlZQ0KMi4gRWZx
IElybHNnDQozLiBxc2R3DQo0LiBxcMrydg0KU2ZnIGegY3NneGR3IGVpayB5cyB3bXlkcj4NCkRzayB4endwIHd2dnkgZWZ2IG9layBz
ZyB3YWwgaW9pIGsgamxhZGguDQpOZmcgc3NqIGxrZSBkeWZ4IGFlc29pZG5ybCB0ZHRlLg0KSWJ5IG9hb2wganJudHcgGt1IEdTciBt
ZmxyIGegamFoZWp3LG0KVnRyYmdtZnkgXJmdyBnbHcgGt1IEdTciBtZmxyIGegamFoZWp3LG0KVnRyYmdtZnkgXJmdyBnbHcgGt1IEdTciBt
cSAXMCBobGV0b2YsIGNnbSB2dHJiZyBlIGZ3eiByZmCuIE9lY3cgaw9pIDQ1IFd2cm1saHMuDQpSeHEgeHp3dWUgcHloIGtnLCBsa2Ug
c29meCBiQWggewZlJ3lwIHduaHga2tmeHcu

Dr. [REDACTED]
Head Director of Mystery

The unknown person is talking about a **Pie recipe** that they made, which is ciphered in some way. We can tell it's Base64 from how it looks. Deciphering it gives us the following.

```
Wo dkxi lzls GSr, cgm kamo gs yww tyo ssddrwzxt mfyueusrrik.  
1. Ilfee  
2. Efq Irlsg  
3. Wsdw  
4. Wrdrv  
Sfg a csgxdw eik ys wmydr.  
Dsk xzwp wvvy efv oek sg wal ioi k jladh.  
Nfg ssj lke dyfx aesoidnrl hdrk.  
Iby oaol jrntw lke GSr mflr a jahejw.  
Vtrbgmfy irfw glw lrp cosx, vayiuo vx afwo 100 gsergw.  
Vo vfrvq 10 hletof, cgm vtrbg e fwz rfg. Oecw ioi 45 wvrmhs.  
Rxq xzwue pyh kg, lke sofx HAh yfe'yp wnhr kkfxw.
```

It looks like a cipher, but we don't know what it is yet. Read the rest of the conversation, we notice that "Ford Pines" also couldn't decipher it, so the unknown person gave him the following hint.

--- Original Message ---

From: [REDACTED]
Sent: Saturday, June 16, 2012 02:23
To: FrPines@Mystery.org
Re: Subject: PIe Recipe

>

I would love to help, but I don't anyone to decipher the recipe.
So I think I'll give you the key in the form of a riddle.

The more of this there is, the less you can see. What is it?

Dr. [REDACTED]
Head Director of Mystery

The hint dictates that the cipher uses a key; a very well-known key cipher is the “**Vigenère Cipher**.” The riddle is “**The more of this there is, the less you can see. What is it?**” The answer is **Darkness**. Trying this as the key, we get the following.

 **DARKNESS**

ABCDEFGHIJKLMNOPQRSTUVWXYZ (26)

To make this PIe, you have to get the following ingredients.

1. Flour
2. Any Fruit
3. Salt
4. Water

And a little bit of sugar.
Mix them well and let it sit for a while.
Now for the most important part.
You will shape the PIe into a square.
Starting from the top left, divide it into 100 pieces.
So every 10 pieces, you start a new row. Bake for 45 Minutes.
And there you go, the best PIe you'll ever taste.

They seem like instructions to make a Pie, but why is it written like that? Read the instructions, it says to prepare the Pie into a square, and cut it into 100 pieces, with the first piece starting from the top left, creating a **10x10 grid**. The PIe is a hint to use the first **100 digits of PI** to create the 10x10 grid. We might need to use this later.

For now, let us take a look at the other text file. It also seems to contain conversation, but it's encoded.

```
--- Original Message ---
From: [REDACTED]
Sent: Tuesday, June 19, 2012 02:15
To: FrPines@Mystery.org
Subject: Leaked Messages

>
1.0-7.7 4.7-0.6 6.0-c 1.8-c 1.8-f 2.8-4.6 8.1-d 5.9-0.2 2.6-4.6 2.2-6.0 0.4-f 8.9-0.2 7.9-9.5 5.7-c
1.9-8.7 5.0-1.3 8.9-6.5 8.6-0.1 1.8-9.2 4.7-e 9.1-b 0.2-4.6 5.9-8.0 4.7-f 1.2-d 6.0-9.0 1.8-f 1.8-e
9.1-9.8 5.1-0.7 1.7-4.5 6.4-5.8 2.6-9.4 8.6-c 1.7-9.0 8.6-2.0 4.7-b 1.8-4.0 1.2-e 9.1-8.2 2.6-6.9 8.6-f
8.9-3.0 5.9-5.1 2.5-4.6 7.9-3.6 1.2-f 7.9-e 5.9-1.8 1.7-0.3 2.1-0.2 5.9-2.4 1.2-6.3 2.1-9.6 0.4-8.7 1.2-
f 4.7-e 2.1-6.1 8.8-e
8.1-1.7 2.1-2.9 1.7-f 4.7-d 8.8-4.8 1.2-e 7.9-f 6.4-5.9 2.6-1.3 1.8-f 9.1-e 2.9-c 7.2-0.7 2.1-7.0
5.9-6.1 1.2-0.5 2.3-6.9 2.1-8.5 6.0-7.1 1.2-9.7 8.9-5.8 0.2-4.6 4.7-9.0 8.6-e 1.8-3.2 5.6-5.7 5.6-9.7
2.1-4.8 8.3-6.8 7.9-7.5 6.0-f 0.4-e 2.8-3.5 9.1-d 7.9-9.0 8.3-9.5 7.9-4.3 4.7-f 6.0-6.5 2.7-6.7 6.4-1.0
7.9-f 2.7-9.4 8.3-8.0 6.0-7.0 6.0-e 0.4-6.8 2.7-0.7 6.0-d 0.4-0.3 6.4-0.9 5.9-8.0 8.6-7.6 1.8-8.9
0.4-7.0 5.5-0.9 5.1-e
```

Looking at it, we can see a pattern of “**Number.Number-Number.Number**” or “**Number.Number-Letter**” and then a space. The numbers are in a range of “**0-9**”, and letters are in a range of “**a-f**”. This looks like hexadecimal, but what does the “.” mean? Looking back at the grid we made using the Ple Recipe, we can deduce that they are **X, Y** coordinates which give us a number to create a hexadecimal pair separated with a space.

We can write Python code to decode this.

Solution Code

```
CIPHERTEXT = """.split()

# The First 100 Digits of PI (NOT including 3.)
PIGRID = [
    [1, 4, 1, 5, 9, 2, 6, 5, 3, 5],
    [8, 9, 7, 9, 3, 2, 3, 8, 4, 6],
    [2, 6, 4, 3, 3, 8, 3, 2, 7, 9],
    [5, 0, 2, 8, 8, 4, 1, 9, 7, 1],
    [6, 9, 3, 9, 9, 3, 7, 5, 1, 0],
    [5, 8, 2, 0, 9, 7, 4, 9, 4, 4],
    [5, 9, 2, 3, 0, 7, 8, 1, 6, 4],
    [0, 6, 2, 8, 6, 2, 0, 8, 9, 9],
    [8, 6, 2, 8, 0, 3, 4, 8, 2, 5],
    [3, 4, 2, 1, 1, 7, 0, 6, 7, 9],
]
```


We will define some variables, like the **CIPHERTEXT** and the **PIGRID**, which we made using the first **100 digits of PI**. We split the ciphertext using spaces to get each hex pair into a list.

```
hex_characters = []

for item in CIPHERTEXT:
    item = item.split("-")
    for hex_representation in item:
        hex_representation = hex_representation.split(".")

        if len(hex_representation) > 1:
            target_num = PIGRID[int(hex_representation[1])][int(hex_representation[0])]
            hex_characters.append(target_num)
        else:
            hex_characters.append(hex_representation[0])
```

We need to create a list of the numbers we get from the X and Y coordinates. We will start by defining a “**hex_characters**” list. After that, we will loop over the items in the ciphertext, splitting it at the “-”. This will give us a small list, which has “**Number.Number**” or “**Number.Letter**”. Now we need each number on its own to use as X and Y coordinates. So, we will split it again at the “.” To get individual numbers (**If it’s a letter, it will not change anything**). We check if the “**hex_representation**” has more than one item; if so, that means it’s a number, otherwise it’s a letter. If it’s a letter, we add it to the “**hex_characters**” list; otherwise, we use the number as X and Y coordinates to get the target number (We loop over the outer lists first and then the inner lists, that’s why the X and Y are flipped).

After getting the target number, we added it to the “**hex_characters**” list.

```
str_hex_characters = [str(item) for item in hex_characters]

hex_string = "".join(str_hex_characters)
plaintext = ""

for i in range(0, len(hex_string), 2):
    hex_pair = hex_string[i : i + 2]
    character = chr(int(hex_pair, 16))
    plaintext += character

print(CIPHERTEXT)
print(plaintext)
```

Now we have a list of integers and characters. We will turn them all into strings and combine them into one long string. After that, we are going to create hex pairs.

Now we can convert the hex pairs into ASCII letters using the **chr()** function in Python. Add the results to a string, and that's it; we have the plain text.

Let's test it on the first line in the text file.

```
['1.0-7.7', '4.7-0.6', '6.0-c', '1.8-c', '1.8-f', '2.8-4.6', '8.1-d', '5.9-0.2', '2.6-4.6', '2.2-6.0', '0.4-f', '8.9-0.2', '7.9-9.5', '5.7-c']  
Hello Mr Ford.
```

We can see that it works. Going through the rest of the conversation, we can see that the password to the **"Archive.7z"** is mentioned in the conversation.

```
I'm affraid so. For the time being.In case anything happens to me.I need you to keep this password with you.  
It unlocks the Archive file. You have to take care of it.Password: D3l1c10us$P13
```

We can use it to unlock the **"Archive.7z"** file. Extracting the file, we are met with a final message and the flag: **CYNX{P1e_encrypt710n}**.

```
If you're reading this, then my suspicions are confirmed.  
You have been monitoring our conversations.  
And you were able to crack our code.  
I have anticipated this, so I moved the archived files to a secure server.  
I didn't expect you to get this far, so I'll give you this.  
  
Flag: CYNX{P13_3ncryp710n}  
  
Until we meet again...
```

Tools Used

1. [CyberChef](#)
2. [Dcode Vigenère Cipher](#)
3. Python 3.14
4. 7zip